Yuhyun Lee

Google Scholar profile; LinkedIn profile; yuhyunl@andrew.cmu.edu

Research Interest: Electrical brain stimulation | Neural prostheses | Neuro rehabilitation

EDUCATION

Ph.D., Carnegie Mellon University (CMU), Biomedical Engineering, Pittsburgh, PA, USA — Present
Advisor: Prof. Pulkit Grover
Thesis: Transanasl electrical stimulation (TnES) for minimally invasive deep brain stimulation

M.E, Daegu Gyeongbuk Institute of Science & Technology (DGIST), Robotics Engineering, Daegu, Korea — Feb. 2020
Advisor: Prof. Sohee Kim
Thesis: Development of EEG recording systems from multiple adult and larval zebrafish

B.E, DGIST, Bachelor of Engineering in School of Undergraduate Studies, Daegu, Korea — Feb. 2018
The first cohort of the undergraduate program
Academic advisor: Prof. Chang-Hun Lee | Research project advisor: Prof. Ji Woong Choi

Summer Session Program, Johns Hopkins University, Baltimore, MD, USA — Jun-Jul. 2015

CAREER OBJECTIVES

My career objective is to develop neural prostheses and translate neural engineering technologies into clinically and commercially viable applications. **My early research focused on designing cost-efficient and non/minimally invasive tools for neural recording (EEG) and electrical stimulation in small animal models (mice & zebrafish)**. **I measured evoked LFP by stimulation invasively (from the cortex to the hypothalamus) using a high-density probe (64 channels)**. Additionally, I participated in projects measuring 1) brain activities using fNIRS from human subjects for neurofeedback and signal classification, 2) EMG by transcranial electrical stimulation in the human motor cortex, and 3) electric fields in a human cadaver head with clinicians. Building on this foundation, I aim to pursue translational research spanning preclinical animal studies and clinical-level investigations to address current challenges in neural prosthetics.

SKILLS AND TECHNIQUES

Biology: Neuro-electrophysiology measurement (LFP, EEG, EMG, fNIRS, MEP) | Brain electrical stimulation (TES in human and ex-vivo setups, TnES in an in-vivo rodent model) | Rodent surgery (in vivo/ex vivo/OVX/Intracardiac perfusion) | Stereotaxic apparatus | PFA fixation/Intracranial perfusion | C-fos | Vibratome | Cell staining/labeling/imaging | Zebrafish maintenance & breeding | Cell attached patch-clamp

Electrophysiology device: Intan RHD (Intan Technologies LLC), Ripple (Ripple Neuro LLC), DS8R/DS5 (Digitimer Ltd), Biopac (Biopac Systems Inc)

Hardware: 3D Printing | Acryl laser cutting | Design and fabrication of microfluidic channels (PDMS base, gold etching, Lift-off technique) | Fabrication of parylene-C based electrodes | FPCB & PCB design

Software: MATLAB (signal processing /basic ML/statistics/data visualization/automatic manipulating system /point source simulation, ROAST (MRI-derived human head model simulation for electric field calculation) | COMSOL Multiphysics (FEM simulations for fluid dynamics and electric fields calculation) | Auto CAD & SolidWorks (2D/3D structure design) | KICAD (PCB design) | FIJI/ImageJ | Python

PUBLICATIONS

1. **Yuhyun Lee**, Vishal Jain, Maysam Chamanzar, Pulkit Grover*, Mats Forssell*, "Transnasal approach as a minimally invasive electrical brain stimulation targeting the arcuate nucleus in the hypothalamus in in vivo mice setup" *(In Preparation)*
2. Mats Forssell, Yuxin Guo, **Yuhyun Lee**, Vishal Jain, Alonso Buitano Tang, Dorian Kusyk, Boyle Cheng, Alexander Whiting, Eric Wang, Alexander Yu, Kora Hughes, Pulkit Grover*, "Transnasal Minimally Invasive Deep Brain Sensing and Stimulation for Mental Disorders" *(Under Review)*
3. **Yuhyun Lee**, Vishal Jain, Yuxin Guo, Maysam Chamanzar, Pulkit Grover*, Mats Forssell*, "Influence of out-of-plane electric fields in an ex vivo model of transcranial electrical stimulation" *(In Revision)*
4. **Yuhyun Lee**, Vishal Jain, Maysamereza Chamanzar, Pulkit Grover*, and Mats Forssell*, "Ex vivo studies of efficacy of DeepFocus: a technique for minimally-invasive deep-brain stimulation" In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1-7, Jul. 2024.
5. Yuxin Guo, Mats Forssell, Dorian Kusyk, Vishal Jain, Issac Swink, Owen Corcoran, **Yuhyun Lee**, Alexancer C. Whiting, Boyle C. Cheng, Pulkit Grover*, "*DeepFocus*: A Transnasal Approach for Optimized Deep Brain Stimulation of Reward Circuit Nodes", *Journal of Neural Engineering*, Feb. 2025.
6. **Yuhyun Lee**, Hee Won Seo, Kyeong Jae Lee, Jae-Won Jang and Sohee Kim*, "A Microfluidic System for Stable and Continuous EEG Monitoring from Multiple Larval Zebrafish", *Sensors*, *20*, 5903, Oct. 2020.

7. **Yuhyun Lee**, Kyeong Jae Lee, Jae-Won Jang, Sang-im Lee, and Sohee Kim*, "An EEG system to detect brain signals from multiple adult zebrafish", *Biosensors and Bioelectronics*, vol. 164, 112315, Sep. 2020. **IF: 12.54; JCR 1%**
8. Yu Jin Woo, Kim Bo Min, Lee Do Young, Kim Seung Hyeon, **Lee Yu Hyun**, Kim Kyung Soo and Choi Ji Woong*, "Classification of fNIRS brain imaging signal in prefrontal cortex for non-invasive portable brain-computer interface," *The Korean Institute of Communications and Information Sciences*, pp. 941-942, Jan. 2018.
9. Lee Do Young, Yu Jin Woo, Kim Bo Min, Kim Seung Hyeon, **Lee Yu Hyun**, Kim Kyung Soo, Lim Sung Ho, and Choi Ji Woong*, "Efficient channel selection of fNIRS based BCI system for mobility enhancement," *The Korean Institute of Communications and Information Sciences*, pp. 918-919, Jun. 2017.

CONFERENCES (FIRST AUTHOR ONLY)

1. **Yuhyun Lee**, Vishal Jain, Pulkit Grover, Mats Forssell*, "Transnasal electrical stimulation (TnES) of the deep brain in an in vivo mouse model", Society of Neuroscience, Society for Neuroscience, Washington D.C., USA, Nov. 14-18, 2026.
2. **Yuhyun Lee**, Vishal Jain, Pulkit Grover, Mats Forssell*, "Transnasal electrical stimulation (TnES) for brain modulation in an in vivo mouse model", 2026 NYC neuromodulation", New York City, NY, USA, Jul.30-Aug.3, 2026.
3. **Yuhyun Lee**, Yuxin Guo, Vishal Jain, Maysamerza Chamanzar, Pulkit Grover, and Mats Forssell*, "Predicting neural responses to *DeepFocus*, a new technique for transnasal, deep-brain electrical stimulation", Society for Neuroscience, Chicago, IL, USA, Oct. 5-9, 2024.
4. **Yuhyun Lee**, Yuxin Guo, Vishal Jain, Dorian Kusyk, Maysamreza Chamanzar, Boyle C. Cheng, Alexander Whiting, Mats Forssell, Pulkit Grover*, "*DeepFocus*: A new transnasal approach for minimally-invasive deep-brain stimulation: Simulation, cadaver, and ex-vivo studies to predict responses in the human head", Carnegie Mellon Forum of Biomedical Engineering, Virtual, Sep. 20, 2024
5. **Yuhyun Lee**, Vishal Jain, Maysamerza Chamanzar, Pulkit Grover* and Mats Forssell*, "Ex vivo studies of efficacy of DeepFocus: a technique for minimally-invasive deep-brain stimulation", *IEEE EMBC,* Orlando, FL, USA, Jul. 15-19, 2024.
6. **Yuhyun Lee**, Vishal Jain, Pulkit Grover and Mats Forssell*, "An ex vivo model for non-invasive brain stimulation utilizing perpendicular planes of electrodes", Society for Neuroscience, Washington D.C., USA, Nov. 11-15, 2023.
7. **Yuhyun Lee** and Sohee Kim*, "A system to record EEG from multiple adult zebrafish," *IEEE EMBC*, Berlin, Germany, Jul. 23-27, 2019.
8. **Yuhyun Lee** and Sohee Kim*, "Recording EEG signals from epilepsy-induced zebrafish treated with anti-epileptic drugs," *International Zebrafish Conference*, Suzhou, China, Jun.12-16, 2019.
9. **Yuhyun Lee** and Sohee Kim*, "Liquid exchangeable microfluidic channels to align zebrafish," *The Korean MEMS Conference (KMEMS)*, Jeju, Korea, Apr. 4-6, 2019

PATENTS

1. Mats Forssell, Yuxin Guo, **Yuhyun Lee**, Vishal Jain, Pulkit Grover, Sinus Implant for Deep Brain Stimulation and Sensing, United States Patent.
2. Mats Forssell, Yuxin Guo, **Yuhyun Lee**, Vishal Jain, Chaitanya Goswami, Ariel Feldman, Neil Ashim Metha, Boyle Cheng, Pulkit Grover, Method for Non-Invasive or Minimally-Invasive Stimulation of Deep Brain Targets, United States Patent, Publication No. 20240416125, Dec. 19. 2024
3. Sohee Kim, **Yuhyun Lee**. FISH BIO SIGNAL MEASURING DEVICE. EU patent, Registration No. 3725219, Jan. 22, 2025
4. Sohee Kim, **Yuhyun Lee**. FISH BIO SIGNAL MEASURING DEVICE. Korean patent, Registration No. 10-2237282, Apr. 01. 2021

AWARDS

- Philip and Marsha Dowd Engineering Seed Fund for Graduate Student Fellowships Pittsburgh, PA, 2024
  - Featured in CMU Engineering News, Dec 20th, 2024
- IEEE EMBC NextGen Scholar Award, IEEE Annual International Conference of Engineering in Medicine and Biology society Orlando, FL, 2024
- Bradford and Diane Smith Graduate Fellowship in Engineering, College of Engineering, CMU Pittsburgh, PA, 2024
  - Featured in CMU BME, Apr 24th, 2024
- CMU Biomedical Engineering Department Head's Fellowship Pittsburgh, PA, 2021
- Best academic conference paper award for undergraduates, The Korean Institute of Communications and Information Sciences Korea, 2018

LANGUAGES

Korean | Native **English** | Professional proficiency **Japanese** | Limited professional proficiency (JLPT N1, most difficult) **Mandarin Chinese** | Elementary proficiency (HSK level 5)

PRESS RELEASE

- Mentorship video featured on the official DGIST channel on YouTube, Jan 3rd ,2024.
- Research featured in *Science Codex*, *EurekAlert*, *Technology Networks*, *YTN Science*, *and many* (1 Broadcast, 6 Paper news, 32 Online news, June 18th to 30th ,2020)

TEACHING & LEADERSHIP EXPERIENCES

Korean graduate student associate | E-board & Leadership position | CMU, PA — Fall 2022-Spring 2026
- Interim president: Rebuilding the organization; Communication with Korean big techs, univs, etc. — Spring 2026
- Led the development and distribution of an onboarding guide for incoming graduate students — Fall 2022-Spring 2026

Korean Class | Teacher & Leader | Carnegie Library of Pittsburgh | Pittsburgh, PA — Summer 2022-Spring 2025
- Teaching Korean language to non-Korean students
- Average students' satisfaction (20 responded) under my leadership: 4.7/5.0 — Summer 2023-Summer 2024

Teaching Assistant | Fundamental of Computational Biomedical Engineering | BME, CMU, PA — Fall 2023/2024
- Official students' overall evaluation: 'A=Excellent'.

Research mentor | Department of Robotics, DGIST, Daegu, Korea — Winter & Summer 2018/Winter 2019
- Reference: Prof. Sohee Kim; two out of six intern students joined the group as graduate students

Teaching Assistant | Brain machine interface | Department of Robotics, DGIST, Daegu, Korea — Fall 2018

Buddy Program | Assisting international students' onboarding | Undergraduate, DGIST, Daegu, Korea — Summer 2017

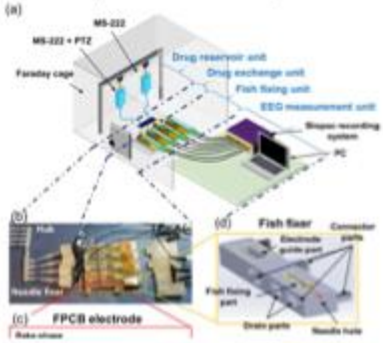
Founder & Leader of Art club | Undergraduate, DGIST, Daegu, Korea — Spring 2015-Summer 2016
- Led fundraising art installation and exhibition development

REFERENCE

Prof. Pulkit Grover | Professor | CMU ECE | pgrover@andrew.cmu.edu | Ph.D. advisor
Dr. Mats Forssell | Research Scientist | CMU ECE | mforssel@andrew.cmu.edu | Ph.D. mentor
Prof. Maysam Chamanzar | Professor | CMU ECE | mchamanz@andrew.cmu.edu | Collaborator
Prof. Sohee Kim | Professor | DGIST Robotics and Mechatronics | soheekim@dgist.ac.kr | Master advisor
Prof. Chang-Hun Lee | Associate Professor | DGIST Department of New Biology | leech@dgist.ac.kr | Undergraduate advisor

RESEARCH SUMMARY

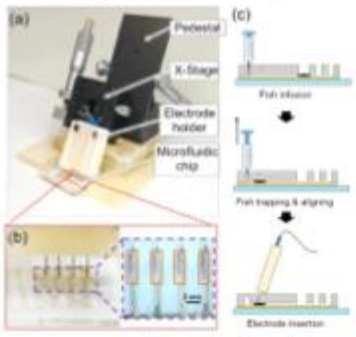
1. Development of EEG system for multiple adult zebrafish



[Goal] For cost-efficient drugs screening for anti-epileptic drugs (AEDs)

[Skills]
- EEG measurements
- Structure design and 3D printing
- FPCB design
- Parylene C- fabrication
- Statistical validation of effects of drugs

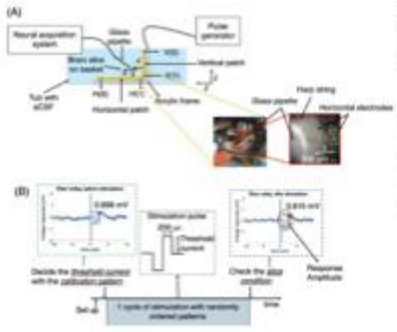
2. Development of EEG system for multiple larval zebrafish



[Goal] For cost-efficient drugs screening for anti-epileptic drugs (AEDs)

[Skills]
- EEG measurements
- Microfluidic channels (PDMS)
- FEM simulation (fluid dynamics)
- Gold deposit & Etching & Lift-off

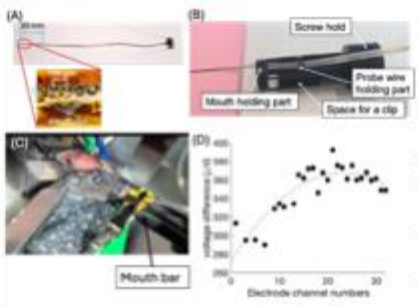
3. Ex vivo experiments for transnasal electrical stimulation (TnES)



[Goal] To present current efficiency of TnES and orientation selective neural responses to stimulation

[Skills]
- Fiber volley measurements
- FEM simulation (electric field)
- Classification (SVM)
- Combine the experimental results with ROAST(electric field calculator using an MRI-derived human head)

4. In vivo experiments for transnasal electrical stimulation (TnES)



[Goal] Proof-of-concept experiment: TnES for deep brain stimulation

- MUA measurement (high density electrodes)
- Structure design
- PCB design for an automatic system
- Intracardiac perfusion fixation
- Probe trace and brain imaging